

Safety and effectiveness of arthroscopic-assisted implantation of allogenic umbilical cord blood-derived mesenchymal stem cells (Cartistem[®])

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Background: The purpose of this study was to investigate the safety and efficacy of arthroscopy-assisted implantation of Cartistem[®], an articular cartilage regeneration treatment composed of allogeneic umbilical cord blood-derived mesenchymal stem cells, during medial owHTO for KL grades II and III patients with medial compartment osteoarthritis of the knee joint; radiological and clinical.

Methods: From October 2017 to July 2019, arthroscopic chondroplasty with Cartistem[®] and medial owHTO were done on lesions with ICRS grade IV in patients with medial compartment arthritis and varus knee. Radiological outcomes including magnetic resonance imaging before and 1 year after surgery were used to calculate the extent of a chondral defect, KL grade, and joint space distance. Furthermore, the KSS and Lysholm score were calculated before and after surgery. The ICRS grade was determined at least 1 year following the operation through the second arthroscopy.

Results: Eight patients who were followed up on for more than 1 year and had the second arthroscopy were investigated retrospectively out of a total of 22 cases. In radiological comparison before and after surgery, the KL grade was improved from III to II in two patients. In 8 patients, the JSD improved dramatically from 3.75 mm to 4.63 mm. The anatomical femorotibial angle was adjusted from varus 2.46° to valgus 9.28°. In the clinical score of 8 patients, the KSS climbed from an average of 46.63 to 70.50 and the Lysholm score increased from an average of 43.5 to 75.5, showing a substantial difference. Furthermore, all patients who underwent a second arthroscopy had a significant improvement in their ICRS grade.

Conclusion: For individuals with KL grade II and III and ICRS Grade IV medial compartment osteoarthritis of the knee, chondroplasty with Cartistem[®] under arthroscopy during medial owHTO can significantly improve the radiological and clinical outcomes.

Keywords: High tibial osteotomy; Arthroscopy; Chondroplasty; Cartistem

INTRODUCTION

Although age is a risk factor for osteoarthritis other factors such as family history, living environment, and lower extremity alignment are also essential [1]. Osteoarthritis affects muscles, ligaments, and the synovial membrane in addition to articular cartilage. Although, improper initial treatment of the joint cartilage defect aggravates osteoarthritis. Therefore, it is significant to effectively treat cartilage defects to avoid osteoarthritis progression [2].

Treatment for articular cartilage defects hinges on the age and activity level of the patient, combined ligament damage or lower limb alignment, and the size and depth of cartilage damage. In cases of wide and progressive cartilage defects and arthritis, arthroplasty is shown. For localized cartilage injury, surgical treatment methods such as microfracture and autologous cartilage grafts are employed [3]. To treat cartilage degeneration in individuals with intermediate cartilage damage, tissue-engineered methods with a scaffold added to autologous cartilage

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cell transplantation are used. The use of scaffolds for tissue-engineering compensates for the drawbacks of first generation autologous cartilage cell transplantation, such as insufficient cross-linking to host cells, incorrect cell delivery, and degeneration of healthy cartilage. To achieve more normal cartilage regeneration, scaffold-based therapy provides a three-dimensional environment for the cartilage lesion, minimizing dedifferentiation of chondrocytes and generating more hyaline cartilage forms [4].

Recently, mesenchymal stem cells (MSCs) have emerged as a new approach in the field of cartilage regeneration, and multiple authors have observed clinical improvement in MSC-treated patients with knee osteoarthritis. MSCs produced from human umbilical cord blood-derived MSCs (hUCB-MSCs) offer significant benefits over other adult autologous stem cells (e.g., bone marrow and adipose tissue-derived MSCs). First, unlike bone marrow-derived MSCs, hUCB-MSCs are acquired non-invasively and do not exhibit an immunological response [5–8]. In addition, because of their great amplification capacity, hUCB-MSCs can provide a significant number of cells for cell therapy. However, at present, there is insufficient evidence for the regeneration of chondrocytes using stem cells [9,10].

It was expected that arthroscopy-assisted transplantation of hUCB-MSCs paired with HTO could result in favorable clinical outcomes in knee osteoarthritis. The goal of this study was to report the safety and efficacy of arthroscopy-assisted transplantation of hUCB-MSCs (Cartistem®; Medipost, Seongnam, Korea), a therapeutic agent for articular cartilage regeneration.

METHODS

In this retrospective study, of the 22 patients who underwent the Cartistem® procedure with high tibial osteotomy from October 2017 to July 2019, the medical results of 8 patients who were followed up for at least 1 year (as outpatients) and underwent a second arthroscopy were reviewed. The 14 individuals who did not undergo the second arthroscopy and followed up less than 1 year were omitted.

The Institutional Review Board (IRB) of the medical records department of Chosun University Hospital records department approved this study (IRB No. 2021-11-005). This study was exempted from informed consent as a study based on medical records. Patients were included if they met the following criteria: (1) over 21 years of age,

(2) they had a more than 5° varus deformity of the knee, (3) they had a cartilage defect size of more than 2.5 cm² on preoperative magnetic resonance imaging (MRI), and (4) they had International Cartilage Repair Society (ICRS) grade 4 cartilage lesions. In addition, individuals with Kellgren–Lawrence (KL) grade II or III osteoarthritis identified using the American college of rheumatology criteria and who had persistent symptoms after conservative treatment for 3 months were included. Patients with knee ligament injury, metabolic arthritis, knee joint infection, and lateral component damage (articular cartilage or meniscus) were excluded from the study (Fig. 1).

Secondary arthroscopy was done to evaluate the cartilage regeneration when the plate for fixation was removed after the bone union.

Preparation of hUCB-MSCs

Cartistem®, an agent for cartilage regeneration, was used in this investigation. It is made up of 1.5 mL hUCB-MSCs (7.5×10^6 cells/vial) and 4% hyaluronic acid (HA) hydrogel, and it was approved by the Korean Food and Drug Administration for cartilage regeneration in January 2012 [11,12].

Surgical procedure

One orthopedic surgeon performed all surgical operations. The degenerative residual cartilage tissue was completely removed through arthroscopic debridement around the lesion where the subchondral bone was exposed due to articular cartilage defects following the diagnostic arthroscopy, synovectomy, and procedure for concomitant injuries such as meniscus injury.

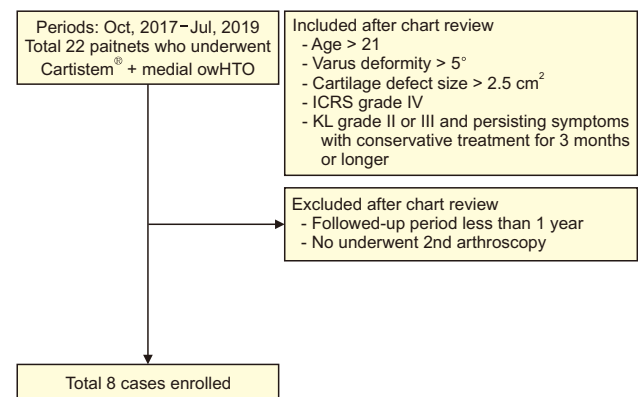


Fig. 1. Flow chart of included patients. KL, Kellgren–Lawrence; ICRS, International Cartilage Repair Society.

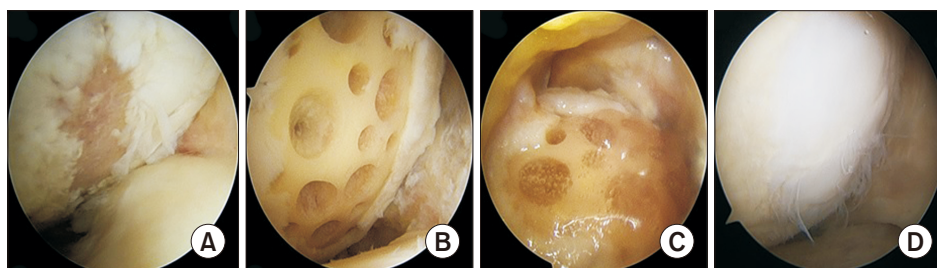


Fig. 2. Arthroscopic findings of stem cell implantation procedures. (A) Medial compartment osteoarthritis. (B) Multiple holes. (C) Human umbilical cord blood-derived mesenchymal stem cell were mixed with hyaluronic acid hydrogel and implanted in the holes. (D) 2nd look arthroscopic image.

To implant the Cartistem[®] using arthroscopy, holes were drilled in the subchondral bone (the cartilage defect area) using a drill with a 4 mm diameter and 4 mm depth; additional holes (using a drill with a 2 mm diameter and 2 mm depth) were made in the area with a sufficient gap between the 4 mm holes. After removing the saline was removed from the knee joint, the joint was inflated with CO₂ injection, and a mixture of hUCB-MSCs and HA hydrogel was applied to the pores and the joint surface (Fig. 2).

Rehabilitation

After surgery, immobilization was achieved by employing long leg splints, and joint motion was permitted on the second day following surgery within pain tolerance, whereas quadriceps strengthening activity was permitted immediately after surgery. Weight-bearing was limited for 6 weeks after surgery, whereas ambulation was accomplished with crutches and wheelchairs; partial weight-bearing was permitted with crutches between 6 weeks and 3 months after surgery. Full weight-bearing ambulation was allowed 3 months after surgery.

Image evaluation

The radiographic alterations from anterior to posterior standing pictures of the knee joint were evaluated for all patients before and after surgery (Table 1). Radiologic data were used to assess KL grade, joint space distance (JSD), and anatomical tibiofemoral (T-F) angle, whereas arthroscopic findings were used to assess ICRS grade before and after surgery.

Clinical outcome evaluation

All patients' clinical outcomes were examined both before and after surgery. The knee society score (KSS) and Lysholm score were used to assess pain, function, range of motion, muscle strength, flexion deformity, joint instability, and joint deformity.

Table 1. Image evaluation outcomes before and after surgery

Evaluation factor	Preoperative	1-year F/U	P-value
KL grade			0.500
I	-	-	
II	3	7	
III	5	1	
JSD (mm)	3.75 ± 0.90	4.63 ± 0.83	0.012
Anatomical T-F angle (°)	-2.46 ± 1.55	9.28 ± 3.64	0.012
ICRS grade			0.010
I	-	4	
II	-	3	
III	-	1	
IV	8	-	

Values are presented as number only or mean ± standard deviation.

-: varus angle, +: valgus angle.

F/U, follow-up; KL, Kellgren–Lawrence; JSD, joint space distance; T-F, tibiofemoral; ICRS, International Cartilage Repair Society.

Statistical analysis

Statistical analysis was done using the Wilcoxon test in the Statistical Product and Service Solution (version 26.0; SPSS Inc., Chicago, IL, USA). The confidence interval was 95%. Significance was set at $P < 0.05$.

RESULTS

The total number of patients was eight (four males and four females); the mean age of the patients was 54.13 years (44–60 y), while the mean body mass index was $28.1 \pm 2.19 \text{ kg/m}^2$. In this research, patients were followed up (as outpatients) for an average of 16.0 ± 3.65 months. On preoperative plain radiography, five and three patients were determined with KL grade II and III, respectively. All eight patients belonged to ICRS grade IV in preoperative MRI, and the mean cartilage defect size was $15.44 \pm 2.62 \text{ mm}$ in the coronal view and 18.94 ± 4.23 in the sagittal view (Table 2).

Regarding KL grade, preoperatively five and three patients had KL II and III, respectively, while seven and one

Table 2. Preoperative evaluation results and dermography for each patient

Case	Age (y)	Sex	BMI (kg/m ²)	Implant removal (mo)	Anatomical T-F angle (°)	KL grade	JSD (mm)	MRI (mm)		Knee society score	Lysholm score
								Coronal gap	Sagittal gap		
1	44	M	29.02	12.36	-1.75	III	4.050	21.86	26.005	36	44
2	60	F	28.98	12.83	-1.40	II	2.815	16.49	21.925	52	34
3	54	F	31.64	12.60	-1.12	III	3.770	15.70	23.425	41	36
4	58	M	28.73	23.50	-5.36	III	2.455	15.85	15.000	23	34
5	50	M	24.30	13.83	-0.84	II	3.790	15.72	20.915	40	40
6	48	M	28.01	13.77	-0.83	III	1.830	14.25	14.105	62	45
7	59	F	28.30	13.30	-3.26	II	2.075	10.98	17.055	68	70
8	60	F	25.96	16.40	-2.74	III	3.105	17.16	20.325	51	50

–: varus angle, +: valgus angle.

BMI, body mass index; T-F, tibiofemoral; KL, Kellgren-Lawrence; JSD, joint space distance; MRI, magnetic resonance imaging.

patient(s) demonstrated KL II and III, respectively, postoperatively. Two patients were enhanced without statistical significance (25.0%, $P = 0.500$).

Analysis of radiographic images indicated that the medial JSD enhanced significantly ($P = 0.012$), from 3.75 ± 0.90 mm preoperatively to 4.63 ± 0.83 mm postoperatively, and the anatomical femorotibial angle also improved significantly ($P = 0.012$), from varus 2.46° to valgus 9.28° .

And the cartilage defect site was reanalyzed employing second arthroscopy at an average of 14.82 ± 3.64 months after the bone union. All eight patients belonged to ICRS grade IV; nevertheless, following surgery, four, three, and one patient(s) represented ICRS grades I, II, and III, respectively, demonstrating significant ($P = 0.010$) improvement. Additionally, the regeneration of tissues similar to the articular cartilage was observed at the site of Cartistem® administration.

Moreover, among clinical parameters, the KSS increased significantly ($P = 0.010$) from an average of 46.63 to 70.50, and the Lysholm score also increased significantly ($P = 0.000$) from an average of 43.50 to 75.50 (Table 3).

DISCUSSION

Microfracture is the most frequently employed technique to repair articular cartilage defects, but the regenerated cartilage is the fibrocartilage and not the hyaline cartilage [13]. In patients treated with this procedure, symptoms aggravate over time (> 2 years), causing articular arthritis problems; furthermore, treatment failure is prevalent in patients with wide defects, obesity, or high activity. Autologous osteochondral transplantation is used as an alternative to this procedure. In this procedure, the patient's

Table 3. Clinical outcomes before and after surgery

Evaluation factor	Preoperative	1-year F/U	P-value
Knee society score	46.63 $\pm$ 14.58	70.50 $\pm$ 8.32	0.010
Lysholm score	43.50 $\pm$ 10.39	75.50 $\pm$ 3.93	< 0.001

Values are presented as mean $\pm$ standard deviation.

F/U, follow-up.

cartilage and bone from a relatively lightweight-bearing area are obtained in the form of a cylindrical plug and transplanted into the cartilage defect site. This procedure is linked to the advantage of regenerating cartilage closest to the hyaline cartilage, but it does not lower the defect at the donor site. Nevertheless, this technique is of limited use when the cartilage lesion is large [14]. Autologous cartilage implantation is another procedure where a small amount of cartilage collected from patients' cartilage is grown in the laboratory and subsequently transplanted into the cartilage defect site at the joint. In the first generation of this technique, cartilage transplantations were conducted employing the periosteum. Scaffolds were utilized in the second generation, whereas the third generation involved the transplantation of cells onto the scaffold [14]. The disadvantage of this procedure is that during culturing of autologous chondrocytes, cell aging progresses, causing an elevation in fibrocartilage components rather than hyaline cartilage. Moreover, in this procedure, two surgical procedures are needed (primary for obtaining cells for culturing and secondary for transplantation) [15]. MSC therapy is an effective alternative treatment technique for osteoarthritis of the knee. MSCs are pluripotent undifferentiated adult stem cells that can be differentiated into osteocytes and chondrocytes and have been expected to regenerate cartilage defects. Recently, it has been de-

liberated that MSCs generate a paracrine effect where the cartilage healing is enhanced by the secretion of several growth factors or cytokines rather than by the direct differentiation of transplanted stem cells into chondrocytes [16].

Collecting bone marrow-derived MSCs, traditionally the most extensively utilized procedure for stem cell harvesting, is a challenging and invasive procedure. Nevertheless, UCB-MSCs can be easily obtained from the umbilical cord discarded at birth, and they are relatively closer to primitive cells compared to other adult stem cells. Henceforth, they denote a different source in the treatment of cartilage damage. These cells can evade the host immune system owing to their immunological properties. Additionally, since human leukocyte antigen is not needed, it can be employed for regenerating cartilage regardless of sex, allergic reaction properties, or blood type [11,17]. Additionally, UCB-MSCs have a higher proliferation potential than MSCs obtained from other tissues. Finally, since these cells are generated “readymade,” they can be employed as required [11].

Park et al. [11] noted a successful clinical outcome of a large cartilage defect in the medial femoral condyle of a 31-year-old female patient using Cartistem®. The authors reported that MSC transplantation demonstrated better results in cartilage regeneration compared to when treated with 4% HA alone. Therefore, Cartistem® is a safe treatment having a low risk of side effects such as rejection and infection and has resulted in high-quality regeneration of cartilage defects. No side effects or serious deviations have been detected even after 7 years of follow-up.

This research has several limitations. First, a biopsy of the regenerated cartilage was not conducted. To accurately examine the state of cartilage regeneration using Cartistem®, it is necessary to visually examine the state of the regenerated cartilage; nevertheless, the space in the knee joint is narrow and the regenerated cartilage can get damaged while using a probe or other tools. These facts restricted biopsy, and thus, the exact regenerated cartilage was not investigated. Second, it was impossible to identify whether the enhancement in clinical symptoms was

owing to chondroplasty using Cartistem® or an additive effect of microfracture and osteotomy. Third, this research was a retrospective analysis without a control group. Furthermore, the number of patients included in the study was also small, there is a selection bias error. A large-scale study is essential to identify the correlation between the clinical trial findings and the safety and effectiveness of arthroscopy-assisted implantation of stem cells. In conclusion, for patients with KL grade II and III and ICRS Grade IV medial compartment osteoarthritis of the knee joint, the safety and efficacy of chondroplasty employing articular cartilage regeneration treatment (Cartistem®; allogenic UCB-MSCs) was verified for the treatment of knee articular cartilage defect regeneration in patients who also received medial open wedge high tibial osteotomy.

As the process significantly enhanced the radiological and clinical outcomes in patients, the clinical application of the arthroscopic regenerative treatment of articular cartilage employing Cartistem® is highly likely. Previously, researchers have reported the clinical use of Cartistem®; nonetheless, clinical results of its use in conjunction with arthroscopic microfracture have not been reported. Later, additional multicenter studies on the application of microfractures using arthroscopy and comparative studies with open surgery during high tibial osteotomy should be performed in the future.

CONFLICT OF INTEREST

No potential conflict of interest relevant to this article was reported.

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